SUPPORTING ADULT GUIDANCE

Stretch and reflect

Quadratics. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

This is the first of two lessons on **transforming** the parabola. It looks only at the **coefficient a** in **$y = ax^2$**. The single idea is that a **scales every height**: multiply the y-value of $y = x^2$ by a. So $a > 1$ gives a **vertical stretch** (a taller, narrower-looking curve), $0 < a < 1$ gives a **compression** (wider, flatter), and $a < 0$ gives a **reflection in the x-axis** (the U becomes a ∩). Crucially, the **turning point (0, 0)** and the **line of symmetry $x = 0$** do not move — only the steepness and the direction change. Sliding the curve sideways with $y = (x - p)^2$ is the *next* lesson; today deliberately keeps the vertex at the origin so a's single job is clear.

Driving Question: *What does the number in front of x^2 do to the curve?*

The mathematics, in plain terms

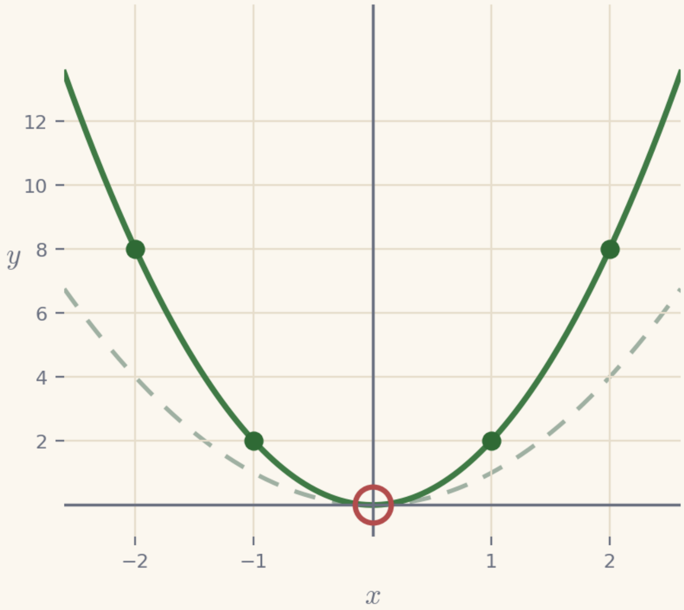
In **$y = ax^2$**, the number **a** is a **multiplier for the height**. Take the ordinary curve $y = x^2$; to get $y = 2x^2$ you double every height, and to get $y = ½x^2$ you halve every height. Doubling the heights makes the curve shoot up faster, so it looks **narrower**; halving them makes it rise slowly, so it looks **wider**. If a is **negative**, every height becomes negative, so the whole curve flips to open **downward** (a ∩ instead of a U). The one thing that never changes is the bottom (or top) of the curve: since $a \times 0^2 = 0$, the **turning point stays at (0, 0)** whatever a is. The most useful support you can give is to keep tying the shape back to the **heights**: "work out $y = x^2$ first, then multiply by a". The most common intuition to correct is the natural guess that a *bigger* number makes a *wider* curve — it is the opposite, because bigger heights mean a steeper climb.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — stretch it — y = 2x^2

Plot y = 2x^2 by scaling y = x^2. For each x, work out x^2 and then multiply by 2. The dashed curve is y = x^2 for comparison.

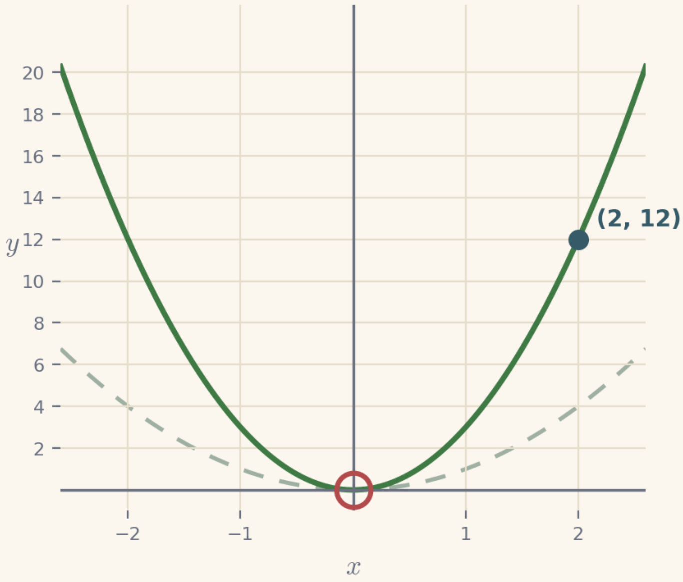


1. x = 1	2 × 1^2 = 2
2. x = 2	2 × 2^2 = 8
3. compare with x^2	every height is doubled

y = 2x^2 is narrower and steeper than y = x^2, but shares the turning point (0, 0)

Example 2 — make it pass through a point

Find a so that $y = ax^2$ passes through (2, 12). Put the point into the rule and solve for a.



1. the rule — $y = ax^2$
2. put in the point — $12 = a \times 2^2$
3. so — $12 = a \times 4$
4. divide — $a = \frac{12}{4} = 3$

a = 3, so $y = 3x^2$ passes through (2, 12)

Scale the height, then read the shape

Two moves carry this lesson. First, **scale the height**: to plot $y = ax^2$ at any x, work out x^2 and then multiply by a — doing this for a couple of points makes "a stretches the curve" concrete rather than a slogan. Second, **read the shape back**: given a curve, ask "is it narrower or wider than $y = x^2$, and which way up?" so the learner links steepness to the *size* of a and direction to the *sign* of a. Keep a still copy of $y = x^2$ in view as a benchmark — every judgement is a comparison with it. Resist introducing side-to-side movement (that is the next lesson); here the vertex stays at the origin so a's effect is isolated. The stretch — finding a so the curve passes through a point — is a first taste of working backwards from data to a rule ($a = y \div x^2$); reach it only once stretch, compression and reflection are secure.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing

Connection	Same curve, scaled: a table for $y = 2x^2$ and $y = ½x^2$, all three plotted together.	~9 min
Movement	The a-dial: move a live control and watch $y = ax^2$ stretch, flatten and flip against $y = x^2$.	~14 min
Reflection	Read the steepness: name the a that produced a drawn curve; catch 'bigger = wider'.	~10 min
Creativity	Make it pass through a point (stretch): find $a = y \div x^2$ so the curve hits a chosen point.	~8 min
Rest	A pause after turning one curve into a whole family.	~2 min
Nutrition	Mode B (intellectual): what the learning feeds — the idea of a transformation, one control governing a family of curves.	~2 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Thinking a bigger a makes a wider curve

Repair: "Check the heights. $y = 3x^2$ at $x = 2$ is 12, but $y = x^2$ is only 4 — bigger heights climb faster, so a bigger a is narrower, not wider."

Believing a moves the turning point

Repair: "Put $x = 0$ in: $a \times 0^2 = 0$ for any a. The turning point stays at (0, 0); only the steepness changes."

Forgetting a negative a flips the curve

Repair: "A negative a makes every height negative, so the curve opens downward — a ∩, not a U."

Multiplying before squaring in ax^2: reading $2x^2$ as $(2x)^2$

Repair: "Square first, then multiply: for $2x^2$ at $x = 3$ it is $2 \times 3^2 = 2 \times 9 = 18$, not $(2 \times 3)^2 = 36$."

(stretch) Not knowing how to find a from a point

Repair: "Put the point into $y = ax^2$. For (2, 12): $12 = a \times 4$, so $a = 12 \div 4 = 3$."

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line

Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson's Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: plotting $y = x^2$ (Lesson 5), multiplying by a whole number and by a half, and squaring including squaring a negative. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Keep the learner scaling heights and comparing with $y = x^2$:

- "Work out $y = 2x^2$ at $x = 1, 2, 3$. How do the heights compare with $y = x^2$?"
- "Is $y = ½x^2$ narrower or wider than $y = x^2$? Why?"
- "Which way up is $y = -3x^2$? How can you tell from the a?"
- "Stretch: what a makes $y = ax^2$ pass through (2, 20)? ($a = 20 \div 4$.)"

If they stall, that is the moment for a Reconnection Route or the lesson's prompt and hint — not for the answer.

Questions you might have

"Why does a bigger a make a narrower curve, not a wider one?"
Because a multiplies the height. Bigger heights mean the curve climbs away from the axis faster, so it looks narrower. It is a very common and natural thing to get the wrong way round at first.

"What stays the same when a changes?"
The turning point (0, 0) and the line of symmetry ($x = 0$). Since $a \times 0^2 = 0$, the vertex never moves — only the steepness and the direction (up or down) change.

"Do they meet $y = (x - p)^2$ in this lesson?"
No — sliding the curve sideways is the next lesson. Keeping the vertex at the origin here means the coefficient a is the only thing changing, so its effect is clear.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Emerging → Developing → Secure → Mastering across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Emerging	Beginning to engage; names some features and makes a start with support.
Developing	Carries out the method with prompts; can explain part of the reasoning.
Secure	Works through independently and explains why each step is true.

Mastering	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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